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Palaeomagnetic data from the Borrowdale Volcanic Group: volcano-tectonics and Late Ordovician palaeolatitudes

J. E. T. CHANNELL[1] & C. McCABE[2]

[1]*Department of Geology, University of Florida, Gainesville, FL 32611, USA*

[2]*Department of Geology and Geophysics, Louisiana State University, Baton Rouge, LA 70803, USA*

Abstract: Nineteen palaeomagnetic sampling sites from the Borrowdale Volcanic Group between Borrowdale and Langdale (English Lake District) yield high unblocking temperature magnetization components which postdate the structural tilt, and therefore postdate the formation of the broad block-faulted syncline which characterizes this part of the central Lake District. The mean magnetization direction before tilt correction (Dec: 347.2°, Inc: −61.9°, α_{95}: 6.9°) corresponds to a palaeolatitude of 43°S (+5°/−7°). The magnetization is associated with pervasive Late Ordovician hydrothermal alteration, and is certainly pre-Devonian based on comparison with available palaeomagnetic data from southern Britain. This implies that the block-faulted synclinal structure of the Borrowdale/Langdale region is pre-Acadian in age, consistent with recent interpretations of Late Ordovician volcano-tectonics based on mapping and volcanological studies.

Palaeozoic palaeogeography of the major continents bordering the Iapetus Ocean has been substantially revised in the last few years, largely as a result of new palaeomagnetic data (see Van der Voo 1988). In Early Ordovician time, the 'Appalachian' margin of Laurentia was at low southerly latitudes and the North African margin of Gondwana at high southerly latitudes, with the south pole in NW Africa. The position of southern Britain (also referred to as Eastern Avalonia) within the Ordovician Iapetus has been the subject of recent debate. Briden *et al.* (1984) considered the latitudinal width of the Llanvirn–Llandeilo Iapetus between Eastern Avalonia and Laurentia to be about 10°. Recent Ordovician (Llanvirn) palaeomagnetic data from the Shelve Inlier (McCabe & Channell 1990) and from the Builth Inlier (McCabe *et al.* 1992; Channell *et al.* 1992*a*) indicate palaeolatitudes of 51°S and 54°S, respectively. High palaeolatitudes of about 62°S are implied by the Arenig Treffgarne volcanic rocks (Dyfed) (Trench *et al.* 1992). Significantly lower Llanvirn palaeolatitudes of about 35°S for Builth, derived from earlier palaeomagnetic data (Briden & Mullan 1984; Trench *et al.* 1991), are interpreted to be an artifact of volcano-tectonic tilting at the sampling sites near the Llanelwedd Quarry or to secular variation of the geomagnetic field not being averaged out at these sites (Channell *et al.* 1992*a*; McCabe *et al.* 1992). The recently-derived higher Ordovician palaeolatitudes for Eastern Avalonia indicate a latitudinal width for the Arenig–Llanvirn Iapetus of at least 30° (3300 km).

The English Lake District has been a focus for Ordovician palaeomagnetic work due to the abundance of volcanic rocks of this age. Briden & Morris (1973) reported palaeomagnetic data from the Eycott Group, and from the Carrock Fell Gabbro Complex. Trench & Torsvik (1991) have revised the structural tilt corrections for these data and have utilized the revised poles for the Llandeilo part of their polar wander path for southern Britain. Also included in this polar wander path are the data of Faller *et al.* (1977) from the Borrowdale Volcanic Group. In the Faller *et al.* (1977) study, a total of 31 sites from the eastern (Ullswater and Haweswater) and SW (Dunnerdale) part of the Borrowdale Volcanic Group outcrop area yielded a well defined mean direction (Declination: 329°, Inclination: −49°, α_{95}: 7°) which passes a fold test indicating the existence of pre-tilt magnetization component.

We report palaeomagnetic results from 19 sites in the central Lake District (Langdale/Borrowdale) region. This area was mapped by Oliver (1961), more recently by Branney (1988 1991) and Davis (1989), and a new map is currently being prepared by Kokelaar *et al.* (University of Liverpool) in association with the British Geological Survey. The region has been the focus of recent revision of the structure and stratigraphy of the Borrowdale Volcanic Group (Branney 1988; Branney & Soper 1988).

Geological setting

The English Lake District lies on the southern side of the Iapetus suture, the suture lying close to the England/Scotland border (Fig. 1). The stratigraphy in the Lake District comprises Tremadoc–Llanvirn marine siliciclastic rocks of the Skiddaw Group unconformably overlain by the Eycott Group in the north, and by the mainly Caradoc Borrowdale Volcanic Group in the south. The Borrowdale Volcanic group is unconformably overlain by shallow marine clastic and carbonate rocks of the Coniston Limestone Formation of the Windermere Group (Moseley 1990). Thick Silurian mudstone–sandstone turbidites and shales lie above the Coniston Limestone Formation and constitute the bulk of the Windermere Group (Moseley 1990).

Deformation within the Borrowdale Volcanic Group and associated with the bounding unconformities has been attributed to caldera-related volcano-tectonics (Branney & Soper 1988), and the main compressive deformation in the area is associated with Acadian orogeny, of Emsian (Early Devonian) age (Soper *et al.* 1987).

The age of the Borrowdale Volcanic Group is stratigraphically constrained between the youngest sediments (late Llanvirn) of the Skiddaw Group and the oldest sediments (Ashgill) of the Windermere Group. The basal sediments of the Windermere Group are Ashgill (Cautleyan) age in the central Lake District, and extend back to Onnian (latest Caradoc) and Longvillian (mid-Caradoc) in the eastern Lake District (Howgill Fell and Cross Fell; Woodcock, pers.

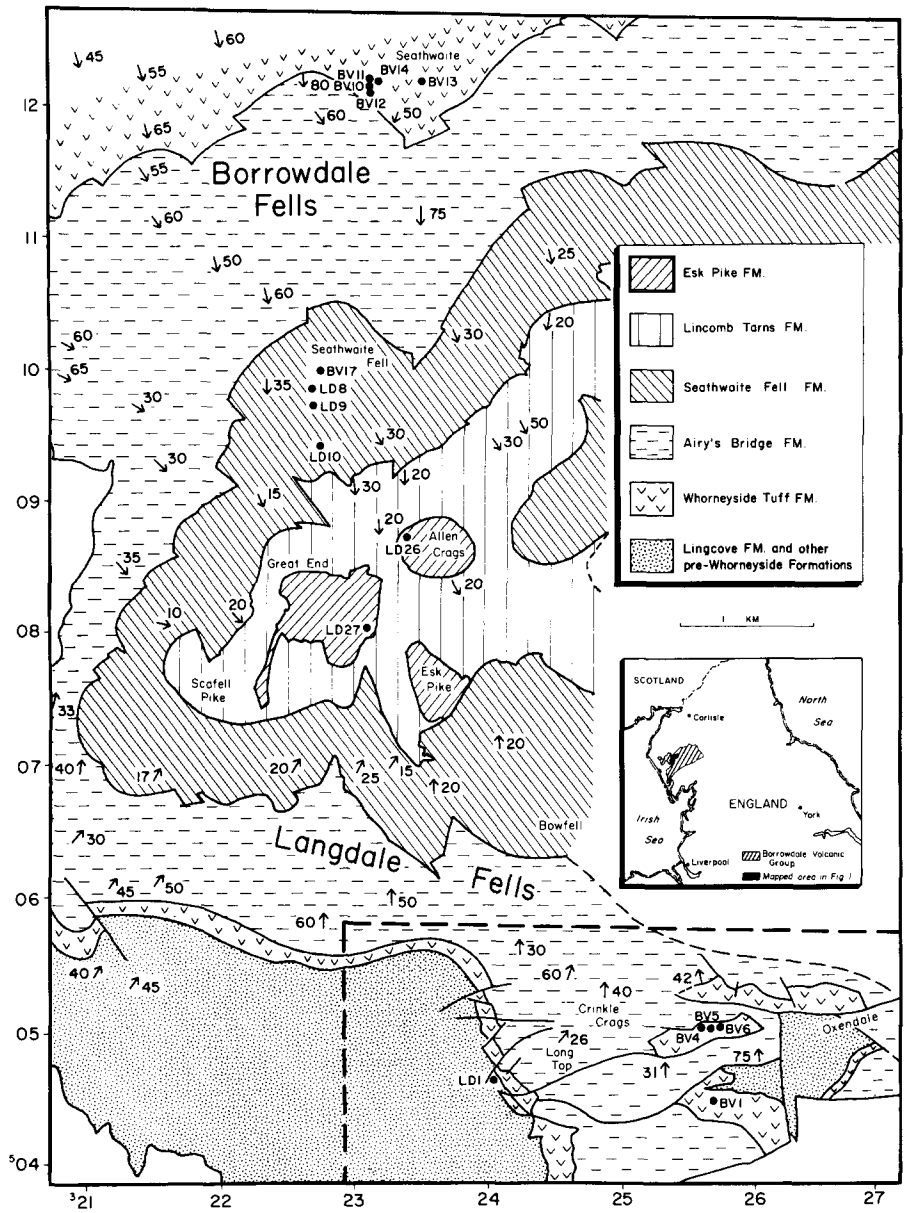


Fig. 1. Geological map of Langdale and Borrowdale after Oliver (1961) and Branney (1991), and after Branney & Soper (1988) for the boxed SE corner. Palaeomagnetic sampling sites which yielded stable magnetizations are marked. For clarity, the abundent close-spaced volcanotectonic faults have been largely omitted.

comm., 1991). Micropalaeontology of a marine mudstone horizon intercalated with the Borrowdale Volcanic Sequence has been interpreted to indicate eruption of most of the sequence in Caradoc time (Branney & Soper 1988; Ruston & Molyneux 1989). This is more or less consistent with a Llandeilo/Caradoc Sm-Nd garnet age of 457 (±4) Ma recorded by Thirlwall & Fitton (1983) from the lower part of the sequence. Eruption and deposition of the Borrowdale Volcanic Group was probably a short-duration geological process, not continuing for many millions of years (Branney, pers. comm., 1989). From available age constraints, we conclude that the Borrowdale Volcanic Group was erupted and deposited in Caradoc time.

The Central Fells area of the Lake District was originally mapped by Oliver (1961). It has since been mapped in more detail (Fig. 1), with revision of the stratigraphy, by Branney (1988), Davis (1989) and Kokelaar *et al.* (work in progress). According to the recent mapping, the Borrowdale Volcanic Group in the Central Fells comprises six formations (Branney 1988, 1991; Davis 1989; Branney & Soper 1988; Branney *et al.* 1992). These are, from the base upwards: (1) Lingcove Formation (andesite and basalt sheets with thin intercalated sediments of similar composition). (2) Whorneyside Tuff Formation (andesitic ignimbrite and phreatomagmatic airfall tuff). (3) Airy's Bridge Formation (silicic welded ignimbrites and subordinate phreatomagmatic tuffs). (4) Seathwaite Fell Formation (lacustrine and fluvial volcaniclastic mudstones, sandstones and breccias, with interstratified pyroclastic rocks). (5) Lincomb Tarns Formation (silicic ignimbrites). (6) Esk Pike Formation (lacustrine volcaniclastic mudstone and siltstone).

In the Central Fells area where the top of the Borrowdale Volcanic Group is absent, the total thickness of the six formations is about 2000 m. In the SW Lake District, the total thickness of the Borrowdale Volcanic Group reaches about 6000 m. The Whorneyside Tuff Formation has a thickness of 120 m at its type locality in Oxendale (NY260052, Fig. 1) and can be traced into Borrowdale, and over a large part of the Borrowdale Volcanic Group outcrop, making it a useful

Table 1. *Palaeomagnetic sampling sites in the Central Fells of the English Lake District*

Site ID	Formation	Grid ref.	S/D
BV1	Whorneyside Tuff Fm.	NY259044	44/9
BV4	Whorneyside Tuff Fm.	NY257051	294/30
BV5	Whorneyside Tuff Fm.	NY258051	296/34
BV6	Whorneyside Tuff Fm.	NY258052	297/33
BV7*	Seathwaite Fell Fm.	NY284073	240/20
BV8*	Seathwaite Fell Fm.	NY282073	244/24
BV9*	Seathwaite Fell Fm.	NY282073	339/43
BV10	Whorneyside Tuff Fm.	NY232121	77/52
BV11	Whorneyside Tuff Fm.	NY232122	76/51
BV12	Whorneyside Tuff Fm.	NY232120	80/51
BV13	Whorneyside Tuff Fm.	NY237122	84/46
BV14	Whorneyside Tuff Fm.	NY233120	92/31
BV17	Seathwaite Fell Fm.	NY229098	74/28
LD1	Lingcove Fm.	NY241046	19/11
LD8	Seathwaite Fell Fm.	NY229098	59/30
LD9	Seathwaite Fell Fm.	NY229096	89/21
LD10	Seathwaite Fell Fm.	NY230093	56/28
LD26	Esk Pike Fm.	NY237087	74/21
LD27	Esk Pike Fm.	NY233078	318/15
Sites with unstable magnetization (not plotted in Fig. 1)			
BV2	Whorneyside Tuff Fm.	NY257045	
BV3	Whorneyside Tuff Fm.	NY261052	
LD2	Lingcove Fm.	NY242047	
LD3	Whorneyside Tuff Fm.	NY243048	
LD11	Seathwaite Fell Fm.	NY237096	

S/D, strike in degrees E of N and dip direction 90° clockwise from strike.

* Located outside Figure 1.

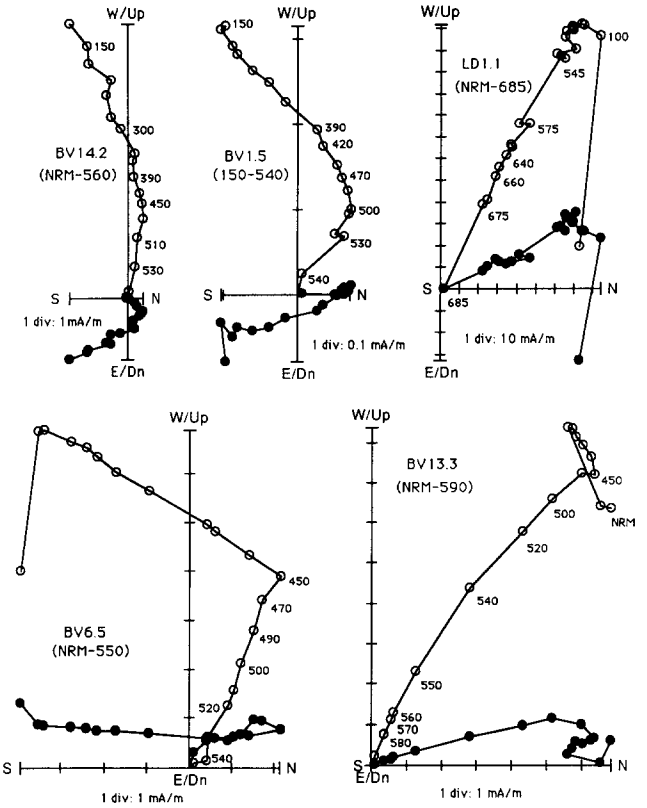


Fig. 2. Orthogonal projection of thermal demagnetization data from the Whorneyside Tuff Formation, with one sample (LD1.1) from the Lingcove Formation. No structural tilt correction. Open and closed symbols represent projection on the vertical and horizontal plane, respectively. Treatment range is given, and temperatures for specific points are given in degrees Celsius. NRM denotes natural remanent magnetization. For key to site numbers see Table 1.

marker horizon in an otherwise complex volcanic pile (Branney & Soper 1988; Branney 1991). The Airy's Bridge Group of Oliver (1961) was renamed the Airy's Bridge Formation and redefined to overlie the Whorneyside Tuff Formation (Branney & Soper 1988; Branney *et al.* 1992).

Palaeomagnetic sampling

A total of 24 palaeomagnetic sampling sites were located around Langdale and Borrowdale in the Central Lake District Fells (Fig. 1). Of the 24 sites (Table 1), 19 sites yielded stable magnetization components. Nine of the 19 sites are in the upper part of the Whorneyside Tuff Formation at the type locality in Oxendale (Langdale area) and at Seathwaite (Borrowdale area) (Fig. 1). The Whorneyside Tuff Formation sampling sites in Oxendale are in the phreatomagmatic airfall tuff, and those at Seathwaite (Fig. 1) are in andesitic airfall tuffs and sediments (Branney 1991). One site (LD1) is in andesite of the Lingcove Formation (Branney 1988, 1991) on the west side of Crinkle Crags (Fig. 1).

Four sites are in volcaniclastic lacustrine mudstones and fine sandstones of the Seathwaite Fell Formation, from Oliver's (1961) type locality on Seathwaite Fell (Fig. 1). Two sites are located in fine-grained volcaniclastic siltsone and mudstone of the Esk Pike Formation (Fig. 1). Three additional sites (with astericks in Table 1) lie within lacustrine volcaniclastic siltstones of the Seathwaite Fell Formation on Harrison Stickle (NY282073).

Palaeomagnetic analysis

Samples were progressively thermally demagnetized at temperature increments of 50°C at low temperatures, with the increment progressively decreasing to 10–25° in the 450–570°C unblocking temperature range. For most of the sites in the Whorneyside Tuff Formation, a distinctive magnetization component, oriented north and up in geographic coordinates, was resolved, usually in the 450–570°C temperature range (Fig. 2). The maximum unblocking temperature suggests that the component is carried by magnetite. For the one site in the andesites of the Lingcove Formation, higher unblocking temperatures indicate a hematite remanence carrier (see LD1.1 in Fig. 2). For the volcaniclastic siltstones and mudstones collected from the Seathwaite Fell Formation and Esk Pike Formation, a well-defined magnetization component was resolved from all but one of the sites analysed (Table 1). The characteristic magnetization component has a north and upward direction in geographic coordinates and was typically resolved in the 450–570°C temperature range (Fig. 3). The directions and unblocking temperature spectra of the characteristic magnetization components from the

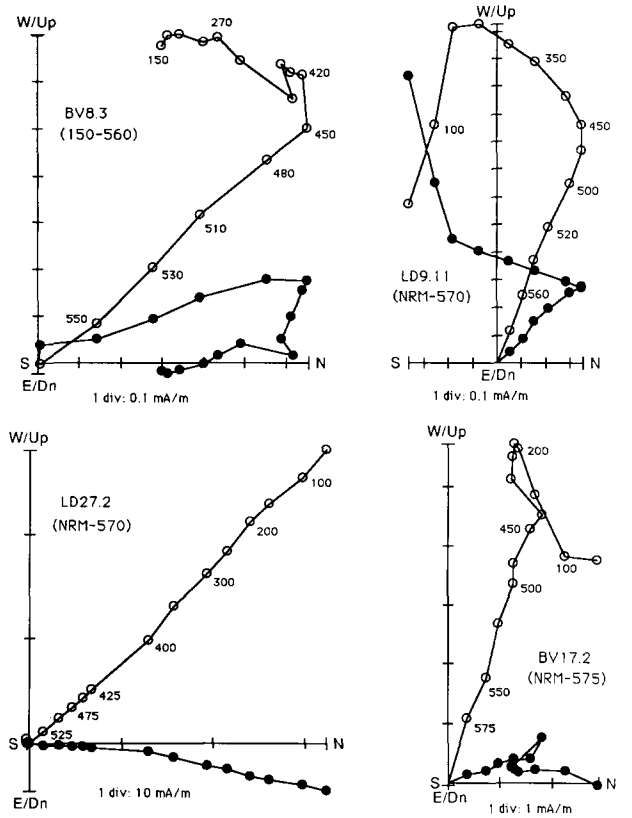


Fig. 3. Orthogonal projection of thermal demagnetization data from the Seathwaite Fell Formation, with one sample (LD27.2) from the Esk Pike Formation. Symbols and notation as for Fig. 2.

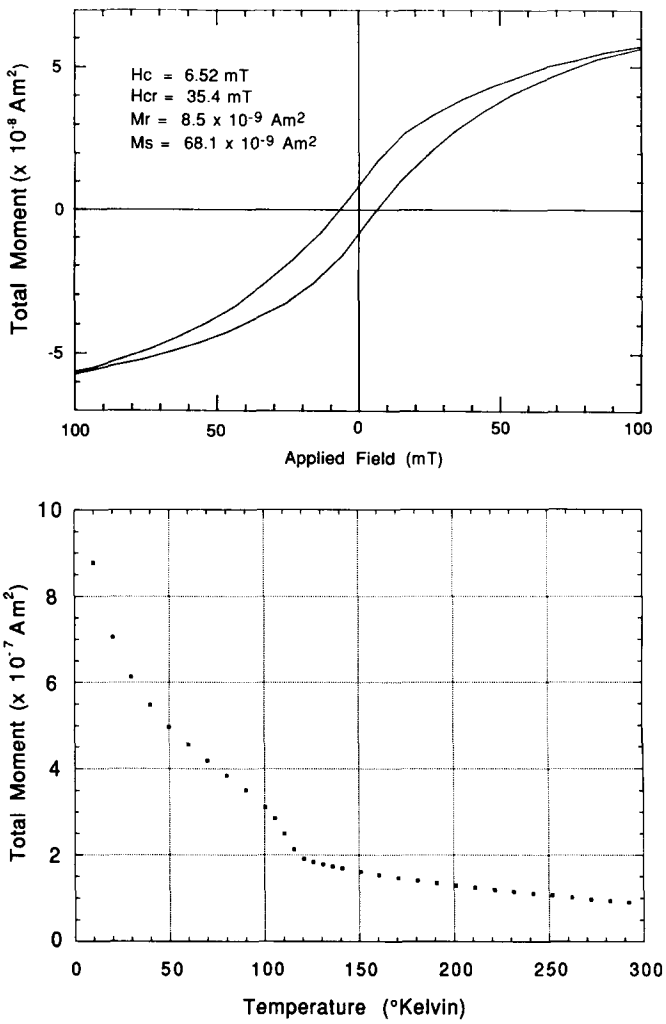


Fig. 4. Hysteresis loop (above) and low temperature remanence loss (below) for a sample of Seathwaite Fell Formation from Site BV17. The hysteresis loop was measured on a Princeton Measurements Corporation Alternating Gradient Force Magnetometer and the low temperature remanence on a Quantum Design MPMS. Hc and Hcr denote coercive force and coercivity of remanence, respectively. Mr and Ms denote saturation remanence and saturation magnetization, respectively. For the low temperature measurements, the sample was magnetized by a 500 mT field at 10°K and the resulting remanence was measured during warming to 300°K.

Whorneyside Tuff Formation and from the Seathwaite Fell Formation are similar.

A lower unblocking temperature magnetization component, with southerly and shallow upward direction, is sometimes resolved in the 100–450°C temperature range both in the Whorneyside Tuff Formation (e.g. BV6.5 and BV13.3 in Fig. 2) and Seathwaite Fell Formation (e.g. BV8.3 and LD9.11 in Fig. 3). The unblocking temperature spectrum of this component often overlaps with the higher unblocking temperature component (e.g. BV1.5 in Fig. 2) and its direction is resolvable in only a few samples at any site. The southerly and shallow upward direction of this component in geographical coordinates suggests that it is Late Carboniferous to Early Permian in age. Remagnetization of this age is ubiquitous in the Welsh Basin (see Channell *et al.* 1992*a*) and in North America (see McCabe & Elmore 1989) and has been associated with remagnetizing fluids emanating from the Variscan deformation front.

Apart from Site LD1, the maximum unblocking temperatures do not exceed 580°C suggesting that magnetite is the main remanence carrier (Figs 2 and 3). Hysteresis parameters (Fig. 4) confirm the dominance of low coercivity remanence carriers and the values of Hrc/Hc and Mr/Ms (where Hrc and Hc denote coercivity of remanence and coercive force, respectively; and Mr and Ms denote saturation remanence and saturation magnetization, respectively) are indicative of multidomain magnetite according to the criteria of Day *et al.* (1977). Warming of a sample carrying a remanence acquired at 10°K indicates a clear discontinuity in remanence decay at 120°K (Fig. 4). This low temperature transition is indicative of magnetite since the magnetocrystalline anistropy constants of magnetite change at this temperature. The presence of the transition also indicates that the magnetite occurs, at least in part, as multidomain grains (Nagata *et al.* 1964; Merrill 1970; Dunlop & Argyle 1991).

The high unblocking temperature component directions were computed from straight line segments on the orthogonal projections of thermal demagnetization data using the standard principal component analysis (Kirschvink 1980). The component directions were then combined for each site using Fisher (1953) statistics (Table 2). Note that a distinctive magnetization component, represented by straight line segments for three or more points on the orthogonal

Table 2. *Site mean directions of the characteristic magnetization component*

Site ID	N/No	k	α_{95}	In situ		Tilt corrected	
				Dec.	Inc.	Dec.	Inc.
BV1	8/10	97.1	5.7	358.1	−48.1	352.1	−41.4
BV4	6/15	116.6	6.2	34.8	−71.6	185.6	−77.5
BV5	6/10	35.1	11.5	355.7	−59.0	272.7	−73.6
BV6	7/14	39.4	9.7	342.3	−71.4	240.3	−67.0
BV7*	7/14	276.9	3.6	340.3	−38.8	345.3	−58.3
BV8*	14/14	106.2	3.9	343.5	−40.7	351.4	−64.0
BV9*	6/13	157.9	5.4	349.1	−50.2	304.0	−39.7
BV10	5/14	66.0	9.5	343.0	−53.6	344.2	−1.6
BV11	14/14	774.8	1.4	352.4	−59.4	349.3	−8.5
BV12	7/14	82.7	6.7	342.4	−54.0	345.1	−3.2
BV13	11/13	36.6	7.6	351.1	−51.0	352.1	−5.0
BV14	11/13	66.1	5.7	114.2	−73.0	34.8	−61.6
BV17	12/13	19.9	10.0	318.6	−64.7	330.6	−38.1
LD1	11/14	124.3	4.1	339.8	−56.0	329.7	−48.3
LD8	12/15	195.9	3.1	345.8	−59.6	338.8	−30.3
LD9	9/12	26.0	10.3	328.8	−60.8	339.9	−41.6
LD10	6/14	71.1	8.0	347.9	−70.0	332.7	−43.4
LD26	12/18	153.7	3.5	336.2	−54.9	338.6	−34.1
LD27	12/16	71.5	5.2	339.2	−47.1	322.3	−50.5
MEAN	16	29.9	6.9	347.2	−61.9		
		6.3	15.9			338.6	−42.8

POLE (based on mean before tilt correction): 8.1°N 6.2°E; dp/dm = 8.2/10.6.

*Volcaniclastic siltstone of the Seathwaite Fell Formation from Harrison Stickle, located outside Fig. 1 and not included in the mean.

N/No is the ratio of samples collected to samples yielding a characteristic magnetization component. k (precision) and α_{95} (circle of 95% confidence) refer to statistical parameters of Fisher (1953). dp/dm represents the oval of 95% confidence about the palaeomagnetic pole. Dec. and Inc. refer to declination and inclination of the site mean directions.

projections, was not resolved from all samples at each site (Table 2). Mixing of unstable demagnetization behavior, with no resolvable magnetization components, and stable behavior were often observed at individual sites.

The site mean directions show significantly better grouping before correction for bedding tilt and become dispersed after tilt correction (Fig. 5, Table 2). The maximum precision of the overall mean is at 0% unfolding, suggesting that the magnetization component was acquired post-tilting, rather than pre- or syn-tilting. The difference in grouping is significant at the 99% confidence level using the criteria of McElhinny (1964) or McFadden (1990). The negative fold test is significant

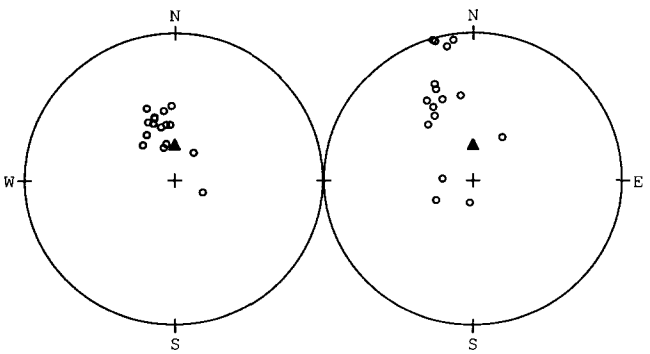


Fig. 5. Equal area projection of all site mean directions from Borrowdale Volcanic Group. Open circles represent upward (negative) inclinations. Closed triangle represents present field direction in the sampling region (downward inclination).

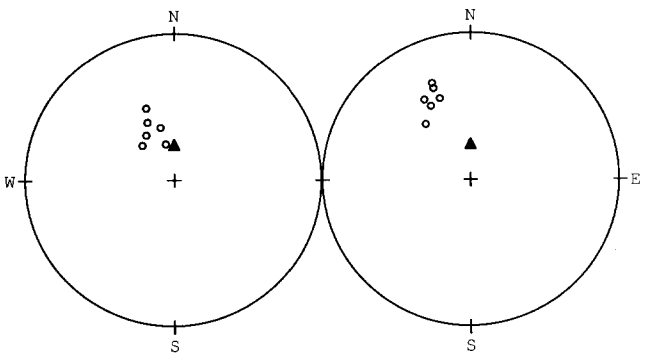


Fig. 6. Equal area projection of site mean directions from Seathwaite Fell Formation and Esk Pike Formation. Symbols as for Fig. 5.

largely due to steep dips at sites from the Whorneyside Tuff Formation, particularly in the Seathwaite area (Fig. 1). For sites from the Seathwaite Fell Formation and Esk Pike Formation, the difference in grouping before and after tilt correction is not significant (Fig. 6, Table 2).

Three sites collected from the Seathwaite Fell Formation on Harrison Stickle are not included in the main group as they are located in the hanging wall of a thrust fault referred to as the Langdale Thrust (Branney *et al.* 1992). Here the compressive deformation is probably Acadian (Branney, pers. comm., 1992). For these sites, the difference in grouping before and after tilt correction is not significant; however, the mean direction is significantly shallower before tilt correction

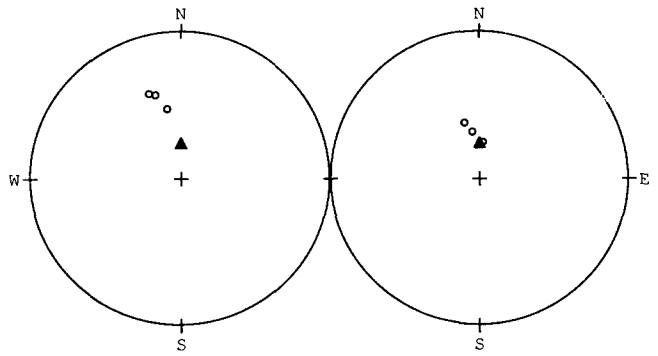


Fig. 7. Equal area projection of site mean directions for three sites from Seathwaite Fell Formation siltstones at Harrison Stickle (NY282073). Symbols as for Fig. 5.

(Fig. 7), and after tilt correction is close to the in situ direction obtained from the other sites (Fig. 5). This suggests that the tilting at the Harrison Stickle locality may be a result of Acadian deformation and that the magnetization predates this deformation.

Interpretation

Post-depositional hydrothermal alteration in the Borrowdale Volcanic Group is indicated by alkali metasomatism and the presence of ubiquitous secondary carbonate, sericite, chlorite and epidote (Branney 1988, 1991). Both the metasomatism and the alteration minerals are typical of late stage hydrothermal alteration in large caldera fields. Alteration is considerably less intense in overlying Ashgill sediments and there is no evidence of major volcanic centres within the Ashgill or Silurian of the main Lake District Inlier (Branney, pers. comm., 1992). We tentatively attribute the characteristic magnetization of the Borrowdale Volcanic Group with a Late Ordovician thermal or thermochemical magnetization process related to broadly syn-volcanic hydrothermal activity prior to the Ashgill onset of Windermere Group deposition. The very similar unblocking temperature spectra of the magnetization component in the Whorneyside Tuff Formation and in the Seathwaite Fell Formation may suggest that the magnetization, carried by magnetite, was acquired by a pervasive (hydrothermal) process affecting much of the volcanic pile. Although the age of our mean magnetization direction is not constrained by a field test (the magnetization is post-tilting), the mean direction and palaeolatitude (43°S) are such that, if the magnetization is single- rather than multi-component, the component is very unlikely to be younger than mid-Silurian in age. By mid-Silurian to Early Devonian time, southern Britain was at much lower latitudes (at about 13° to 17°S) (Channell *et al.* 1992*b*, Torsvik & Trench 1992).

Geological evidence for volcano-tectonics has been clearly documented in Langdale by Branney & Soper (1988). Our negative fold test implies that little, if any, of the structure of the block-faulted 'Scafell syncline' is Acadian in origin, and that the structure is entirely associated with Late Ordovician (pre-Ashgill) volcano-tectonics. This is consistent with the generally poor cleavage development and the apparent absence of compressional buckle folds in this area (Branney & Soper 1988). As explained above, for the site mean directions from Harrison Stickle, there is some indirect evidence that these sediments were tilted by Acadian deformation after they acquired their magnetization.

In contrast to our negative fold test, the data of Faller *et al.* (1977) from Dunnerdale and Ullswater/Haweswater yielded a positive fold test and a mean inclination of −49° (palaeolatitude: 30°S). The positive fold test is due to the predominance of Acadian-aged structures in the Faller *et al.* sampling areas, which lie partly within and adjacent to the zone of intense penetrative deformation on the southern edge of the outcrop area of the Borrowdale Volcanic Group. The small but significant inclination discrepancy between our overall mean direction and the Faller *et al.* (1977) result is difficult to explain, in part because the earlier study was based on blanket alternating field demagnetization treatments. Although blanket treatment of samples was standard procedure at the time of their investigation, the method precludes

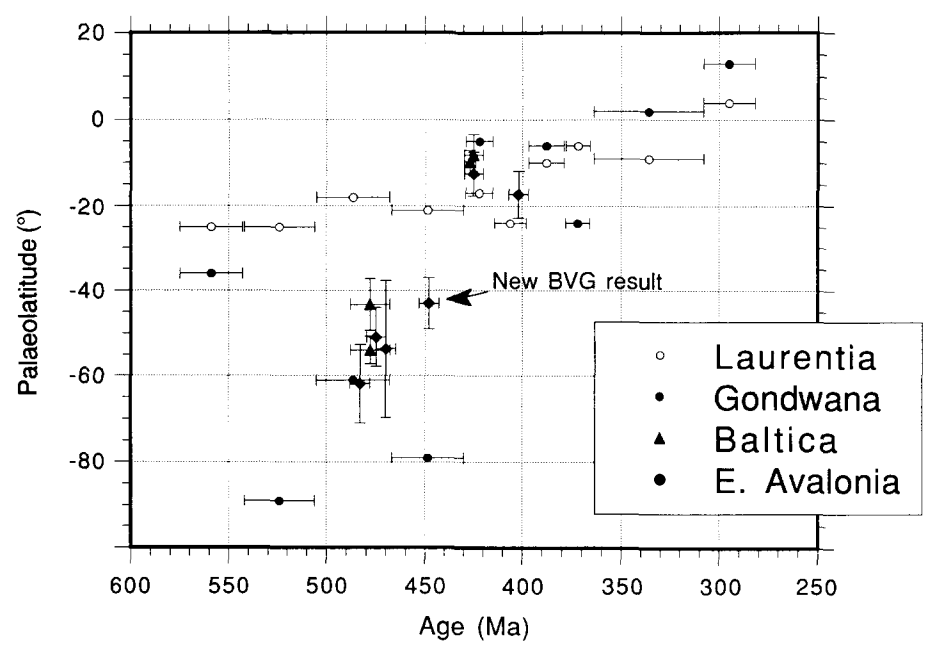


Fig. 8. Predicted palaeolatitudes for southern Britain (after Van der Voo 1992) on the assumption that southern Britain was part of Laurentia (open circles) and part of Gondwana (closed circles), compared with palaeolatitude data from Baltica (Claesson 1978; Douglass 1988; Torsvik & Trench 1991; Trench & Torsvik 1992) and from Eastern Avalonia (southern Britain) (McCabe & Channell 1990; Trench *et al.* 1992; Torsvik & Trench 1992; Channell *et al.* 1992*b*; McCabe *et al.* 1992). The new result from the Borrowdale Volcanic Group (BVG) is indicated.

Table 3. *Key to Figure 9: selected Palaeozoic poles for Eastern Avalonia (southern Britain)*

	Strat. age	Pole Lat (°)	Pole Long (°)	dp/dm
1 Treffgarne Volcanic rocks (Trench *et al.* 1992)	Early. Ord.	56N	306E	9/10
2 Stapeley Volcanic rocks (McCabe & Channell, 1990)	mid. Ord	26.6 N	36.1 E	6.9/8.2
3 Northern Builth (McCabe *et al.* 1992)	mid.Ord	17.9N	12.9E	14.7/17.2
4 Tramore Volcanic rocks (Deutsch, 1980)	mid. Ord	11 N	18 E	10/13
STAR Borrowdale Volcanic Group (this paper)	Late. Ord	8.1 N	6.2 E	8.2/10.6
5 Somerset/Glouc. volcanic rocks (Piper, 1975)	Silurian	8 N	309 E	12.2/17.2
6 Old Red Sandstone (Chamalaun and Creer, 1964)	L. Sil/E.Dev	3 N	298 E	9.2/15.1
7 Old Red Sandstone (Channell *et al.* 1992b)	L. Sil/E.Dev	7.3S	306.7E	5.4/9.5
8 Portishead Red Beds (Morris *et al.*, 1973)	E. Carb	32 S	338 E	5/10
9 Old Red Sandstone overprint (Channell *et al.* 1992*b*)	L. Carb/E. Perm	40.4 S	338 E	3.8/7.5
10 Builth overprint (Trench *et al.* 1991)	L. Carb/E. Perm	44 S	351 E	4/8
11 Builth overprint (McCabe *et al.* 1992)	L. Carb/E. Perm	47.7 S	338 E	3.3/6.4
12 Exeter Lavas (Zijderveld 1967)	E. Perm	50 S	329 E	2/4.3

dp/dm represents the oval of 95% confidence about the palaeomagnetic pole. Prefixes E. and L. refer to Early and Late, respectively.

demonstration of a single component remanence. Similar uncertainty surrounds palaeomagnetic data from the Kentmere area in the southern part of the outcrop region of the Borrowdale Volcanic Group (Piper 1979) and data from Late Ordovician volcanic and plutonic rocks from north Wales (Thomas & Briden 1976).

We tentatively conclude that the high unblocking temperature component in the Borrowdale Volcanic Group predates the Ashgill onset of Windermere Group sedimentation, and has an age close to the Caradoc/Ashgill boundary at about 448 Ma. The overall mean inclination ($-61.9°$, α_{95}: 6.9°) indicates a palaeolatitude of 43°S (+5°/–7°). A Late Ordovician (448 Ma) palaeolatitude of 43°S for Eastern Avalonia is close to the latest Ordovician palaeolatitude of 42°S for Western Avalonia derived from Nova Scotia (Van der Voo & Johnson 1985). The result is also consistent with other data from southern Britain; lying between higher palaeolatitudes for the Arenig and Llanvirn, and lower palaeolatitudes for the mid-Silurian and Early Devonian (Fig. 8). The palaeolatitude from the Borrowdale Volcanic Group is lower than palaeolatitudes recently derived from Early and Middle Ordovician rocks from Wales, where Arenig volcanic rocks at Treffgarne (Trench *et al.* 1992) and Llanvirn rocks at Shelve and Builth (McCabe & Channell 1990, McCabe *et al.* 1992; Channell *et al.* 1992*a*) have yielded palaeolatitude estimates of 62°S, 51°S and 54°S, respectively. Low palaeolatitudes of 13°S and 17°S for southern Britain are implied by mid-Silurian and Early Devonian palaeomagnetic data (Channell *et al.* 1992*b*, Torsvik & Trench 1992), implying closure of the Iapetus Ocean by this time. Comparison of Palaeozoic palaeolatitudes for Eastern Avalonia (southern Britain) with those computed for southern Britain assuming

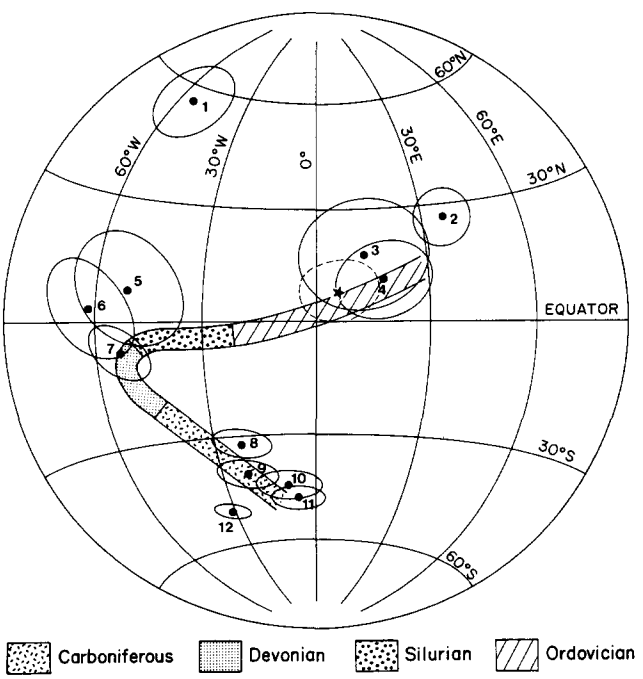


Fig. 9. Selected Palaeozoic poles for southern Britain (Eastern Avalonia) including the new pole from the Borrowdale Volcanic Group (star with dashed 95% oval of confidence). For key to numbered poles see Table 3. The apparent polar wander path for southern Britain computed by Trench & Torsvik (1991) is shown for comparison. Equal area projection.

that this region was part of Laurentia or part of Gondwana (Fig. 8) imply that Eastern Avalonia was a peri-Gondwanide continental unit during Ordovician closure of the Iapetus Ocean.

The new palaeomagnetic pole from the Borrowdale Volcanic Group lies close to recently-derived Mid-Ordovician poles from Wales, and to the Ordovician portion of the apparent polar wander path for southern Britain published by Trench & Torsvik (1991) (Fig. 9, Table 3).

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